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Summary

AI Engineer building **LLM agents, RAG and NLP systems** and the **full-stack platforms** around them, from fine-tuned models to production deployments with **Docker, PostgreSQL and AWS**. Recently shipped an AI agents platform with an **MCP server** and **442 automated tests**, plus client applications for e-commerce and real-estate businesses in the Netherlands.

Experience

AI Engineer **June 2026 – Present**
Optivaize *Netherlands (Remote)*

- Lead developer (**80+ commits**, 21 build phases) of a client **AI agents platform** (**React/TypeScript, Express, PostgreSQL, Docker**) with a **Model Context Protocol (MCP) server** exposing **19 tools** so AI agents can read and manage platform data, plus per-run cost monitoring and automated value reports.
- Delivered **442 automated tests** (Vitest + Playwright) passing on both SQLite and PostgreSQL, with admin MFA, an audit trail and **encrypted backup/restore**.
- Extended and deployed a **generative AI fashion design studio** (**Django, Celery, Redis, PostgreSQL**) using **Gemini** image models with **automatic OpenAI fallback**; shipped region-preserving image edits and job retries, released beta → production on **AWS EC2**.
- Sole developer of a live **sales-plan generator** for a Dutch real-estate agency that pulls property data from **5 government sources** (BAG, Kadaster, CBS, PDOK, EP-Online) and renders shareable, e-signable proposals (**Flask, Docker**).
- For a Shopify retailer, built an **AI image-to-3D pipeline** publishing **540+ 3D/AR models** (**~80%** smaller files, Three.js showroom and garden planner), an **ERP-to-Shopify sync** (**FastAPI**, AWS, **3 stores**), and enriched **830+ products** with **2,300+ verified spec values**.

Associate Engineer (AI) **May 2025 – Feb 2026**
Capsitech *Jodhpur, Raj.*

- Engineered **an NLP-driven assistant** for internal documentation search, cutting lookup **time by 40%**.
- Led integration of **AI microservices** into existing platforms using **Docker** and **RESTful APIs**, ensuring seamless production rollout.
- Implemented modular AI microservices with **Docker, FastAPI** and **Kubernetes**, enabling scalable, low-latency inference (**<200ms**).

AI Intern **Feb 2025 – May 2025**
Siimteq Technologies *Vadodara, Guj.*

- Developed a custom chatbot using **fine-tuned BERT with attention masking**, achieving **91% precision** in intent classification across **50+ user intents**.
- Reduced model response **time by 40%** by implementing **asynchronous inference** and **response caching with Redis** in the FastAPI backend.
- Enhanced **chatbot flow** using custom **dialog states, confidence thresholds**, and fallback strategies to improve response quality.

Projects

Drone Security Analyst Agent [GitHub]

- Built an AI surveillance agent that analyses real drone footage with a **BLIP vision-language model** (dual-pass captioning + **7 safety checks per frame**) and raises alerts from **9 YAML rule types** (loitering, crowding, person down).
- Natural-language querying through a **Groq qwen3-32b agent with 5 tools** over a SQLite frame index; **FastAPI + OpenCV**, with a frame cache that skips the 3–5 s/frame CPU inference on repeat runs.

Conversation Evaluator [GitHub]

- LLM-as-judge pipeline scoring conversation turns on **399 facets** (quality, pragmatics, safety, emotion) with **Llama 3.1**: context extraction, then batched JSON scoring (**25 facets/batch**) with confidence; Streamlit + Docker.

SocietyOps – Society Management Platform

- Built a **multi-tenant web application** for housing-society managers covering **billing, a double-entry accounting ledger** and resident management, using **FastAPI, PostgreSQL** and **React/TypeScript**.
- Exact money handling (integer paise), **maker-checker approvals**, **5 roles** with per-society data isolation; **300+ backend tests** plus **Playwright** end-to-end tests.

Document Analyzer [GitHub]

- Parsed **~120 scientific PDFs** (~12,000 sentences) with BERT and spaCy NER, detecting **900+ mathematical expressions** at **~89% accuracy**; combined **BERT extractive** and **LLaMA 3.3** abstractive summarization for **40–55%** compression.

Education

Parul University **April 2025**
Bachelor of Technology, Computer Science, Specialization in AI & ML *Vadodara, Guj.*

- Coursework: Machine Learning, Data Structures and Algorithms, Artificial Intelligence, Deep Learning, NLP

GATE DS & AI Qualified **May 2025, March 2026**

Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS

AI & Agents: LLM agents, tool calling, Model Context Protocol (MCP), RAG, LangChain, OpenAI / Anthropic (Claude) / Gemini / Groq APIs, Prompt Engineering, LLM evaluation

AI-Assisted Development: Hermes Agent, Claude Code (phased briefs, multi-agent builds, automated test verification)

ML / NLP / Vision: PyTorch, TensorFlow, Hugging Face Transformers, Scikit-Learn, XGBoost, spaCy, NLTK, OpenCV, BLIP, Pandas, NumPy

Web & Backend: React, Vite, Tailwind/shadcn, Node.js, Express, FastAPI, Django/DRF, Flask, Celery, Three.js, Streamlit, Shopify (GraphQL Admin API), Stripe

Data & DevOps: PostgreSQL, SQLite, Redis, Vector databases, Docker, Kubernetes, AWS (EC2), Cloudflare (R2, Tunnel), Nginx, Git/GitHub, Vitest, Playwright, pytest